Benjamin Emag

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LinkedIn · Portfolio Website

Software engineer with a game programming background, focused on systems design, performance optimization, and maintainable C#/.NET and C++ code

EDUCATION

B.S. Game Programming, Minor in Computer Science May 2026
Champlain College *Burlington, VT*

Dean's List: Spring 2023 – Spring 2026
Relevant Coursework: Software Design Patterns, Software Refactoring, Data Structures & Algorithms, Database Management, Computer Architecture, Game Architecture, Advanced Game AI

TECHNICAL SKILLS

Languages: C#, C++, Java, Python, SQL, HLSL/GLSL
Frameworks & Engines: .NET (MVVM), Unity, Unreal Engine 5, OpenGL, Unity ML-Agents
Tools & Process: Git, SVN, Jira, Confluence, Agile/Scrum
Specializations: Software Architecture, Performance Optimization, Simulation, Gameplay Systems

EXPERIENCE

Leahy Center for Digital Forensics & Cybersecurity Burlington, VT
Software Engineer Intern *Summer 2023*

- Developed and maintained .NET applications in C# using MVVM architecture
- Implemented new features and refactored existing code to improve modularity and maintainability
- Debugged and resolved issues across UI and application logic layers in a shared codebase
- Collaborated with researchers and engineers to translate requirements into internal tools for government clients

PROJECTS

Programmer – Fincremental (Capstone) 09/25 – 05/26
Unity, C#, Incremental OS Simulator, Team of 12

- Architected core systems for OS simulation, including window management, upgrades, and persistent saves
- Developed a modular mini-game framework that lets designers easily integrate new content
- Applied SOLID principles throughout the codebase to keep it maintainable and extensible

Solo Developer – Monte Carlo Poker Evaluator Fall 2024
C++, Algorithms/Simulation, Personal Project

- Built a SIMD-accelerated hand evaluator (SSE/AVX), achieving a 1.5x speedup over the scalar baseline
- Applied low-level CPU optimizations to run millions of Monte Carlo simulations efficiently

Solo Developer – Autonomous Flight Simulator Fall 2024
Unity, C#, ML-Agents, Simulation, Personal Project

- Engineered custom flight physics using composite colliders to simulate aircraft dynamics
- Trained a neural network with Unity ML-Agents to perform autonomous takeoff and landing

Programmer – P.I.N.G 01/25 – 05/25
Unreal Engine 5, First-Person Horror, Team of 12

- Developed a custom echolocation shader with UV-independent rendering for seamless use on any mesh
- Engineered a shadow masking system to exclude shadowed geometry from the visualization
- Collaborated with designers to tune visual parameters for gameplay readability

Programmer – Moai Mayhem 01/24 – 05/24
Unity, C#, Educational Adventure, Team of 6, Published on Itch.io

- Implemented the player movement system, integrated with rigged 2D character animations
- Designed and placed collectible Moai statues throughout game environments